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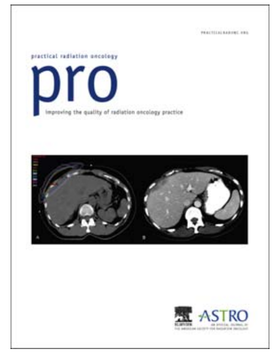

Dose Dissonance in Radiation Oncology: Consensus Needed When Prescribing Dose in Radiation Therapy

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Dose Dissonance in Radiation Oncology: Consensus Needed When Prescribing Dose in Radiation Therapy

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In the current issue of Practical Radiation Oncology Das and co-investigators investigate dose prescription and compliance to international standard ICRU-83 in patients treated with IMRT (1). ICRU-83 was published in 2010 as an international standard for the prescription of IMRT (2). The purpose of international guidelines was to achieve near uniformity in national and international clinical trials to better compare radiation outcomes data. In this study 10 of 36 invited academic medical centers in the United States volunteered to participate. The analysis includes approximately 500 patients from each of the participating institutions for a total of nearly 5100 patients.

Analyzed data included treatment site, technique, planner, physician, prescribed dose, target volume, and dose calculation algorithm. The dose-volume histogram (DVH) of each patient, as well as dose points and parameters as defined in ICRU 83, D100, D98, D95, D50, D2 were collected and analyzed. Additionally, homogeneity index (HI), a measure of the shape of the DVH, was calculated for every patient and analyzed.

The results of the analysis revealed that compliance with ICRU 83 recommendations for naming the target, reporting dose prescription, and achieving desired levels of dose to target were very low. There was statistically significant variability in D95, D50 and HI among institutions, tumor site and technique. Nearly 95% of patients had D50 higher than 100% (103.5±6.9) of prescribed dose and varied among institutions. On the other hand, D95 was close to 100% (97.1 ± 9.4) of prescribed dose. Liver and lung sites had a higher D50 compared to other sites. Variability in D50 is 101.2±8.5, 103.4±6.8, 103.4±8.2 and 109.5±11.5 for IMRT, Tomotherapy, VMAT and SBRT with IMRT, respectively. Pelvic sites had a lower variability indicated by HI (0.13±1.21).

Guidelines from two professional societies have raised concerns about the documentation and implementation of IMRT (3,4). Realizing the consequences of poor dose prescription and delivery, ICRU-83 was published 2010 (1) to provide additional prescription and reporting guidelines in the era of IMRT. The majority (nearly 95%) of patient treatments deviated significantly from the ICRU-83 recommended D50 prescription dose delivery. This variability is statistically significant in terms of treatment site, technique and institution. Others have pointed out the need for consensus when prescribing radiation. For example Eaton and colleagues reported the need for such guidelines when prescribing stereotactic body radiation therapy for prostate cancer (5). In a multicenter study Giglioli et al. showed very large variability in prescriptions (6). Similarly in liver SBRT , the multicenter dosimetry study showed significant variability in prescription and dose recording (7).

It is important to resolve this challenge of dissonance of dose in radiation oncology. Particularly in the era of “big data.” Doing so will permit us to more adequately address issues like those being evaluated by quantitative analysis of normal tissue effects in the clinic (QUANTEC) (8) and efforts using genetics like STROGAR (9) to identify cancer patients at risk for development of adverse outcomes following radiation therapy. To reduce dosimetric and associated radiation outcome variability, dose prescription in every clinical trial should be unified with international guidelines. Replace dissonance with harmony regarding dose in radiation oncology.

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